ABSTRACT

IoT enabled robot for detecting gas pipe leakage

A pipeline traversing robot (100) for real-time gas leakage detection is disclosed. The robot is configured to traverse the exterior surface of metallic gas pipelines in both horizontal and vertical orientations using a three-armed symmetric base (15) carrying wheels (10) and neodymium magnets (3) that maintain continuous radial contact and magnetic adhesion with the pipeline. A processing unit (1) receives data from a gas sensor (2) and a temperature and humidity sensor (6) and processes the data to detect gas leakage. The castor wheels (9) assist the robot to climb pipes vertically and overcome obstacles. An ultrasonic sensor (5) detects obstacles, while an imaging device (4) and a torch (14) facilitate remote visual inspection in low-light environments. Inspection data, robot location, and gas leakage alerts are transmitted to a remote monitoring and control center (12) through an IoT-enabled communication system, enabling real-time monitoring, obstacle-assisted navigation, and timely maintenance of industrial gas pipeline networks.

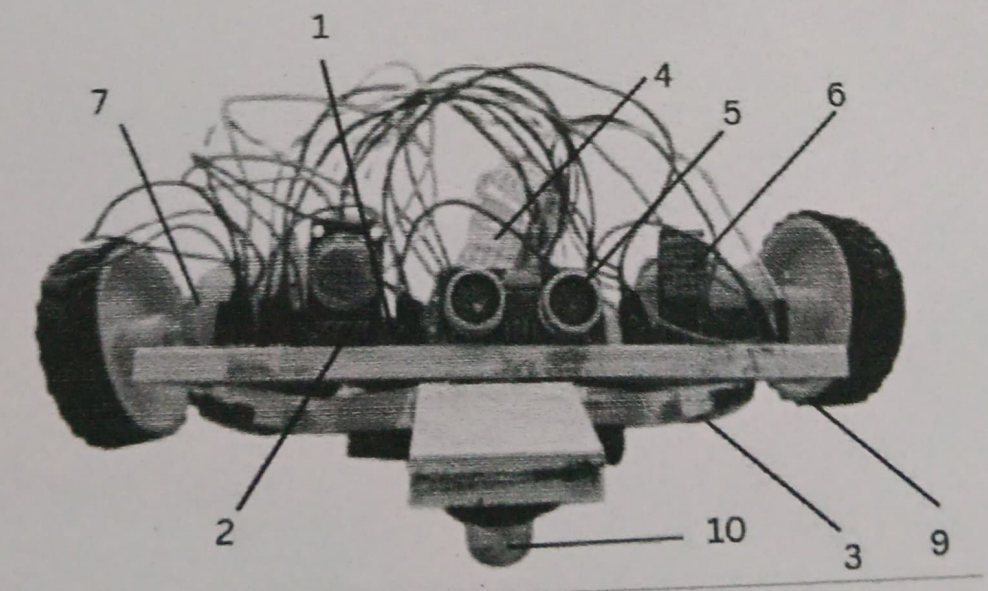


Figure 1

Figure 1

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